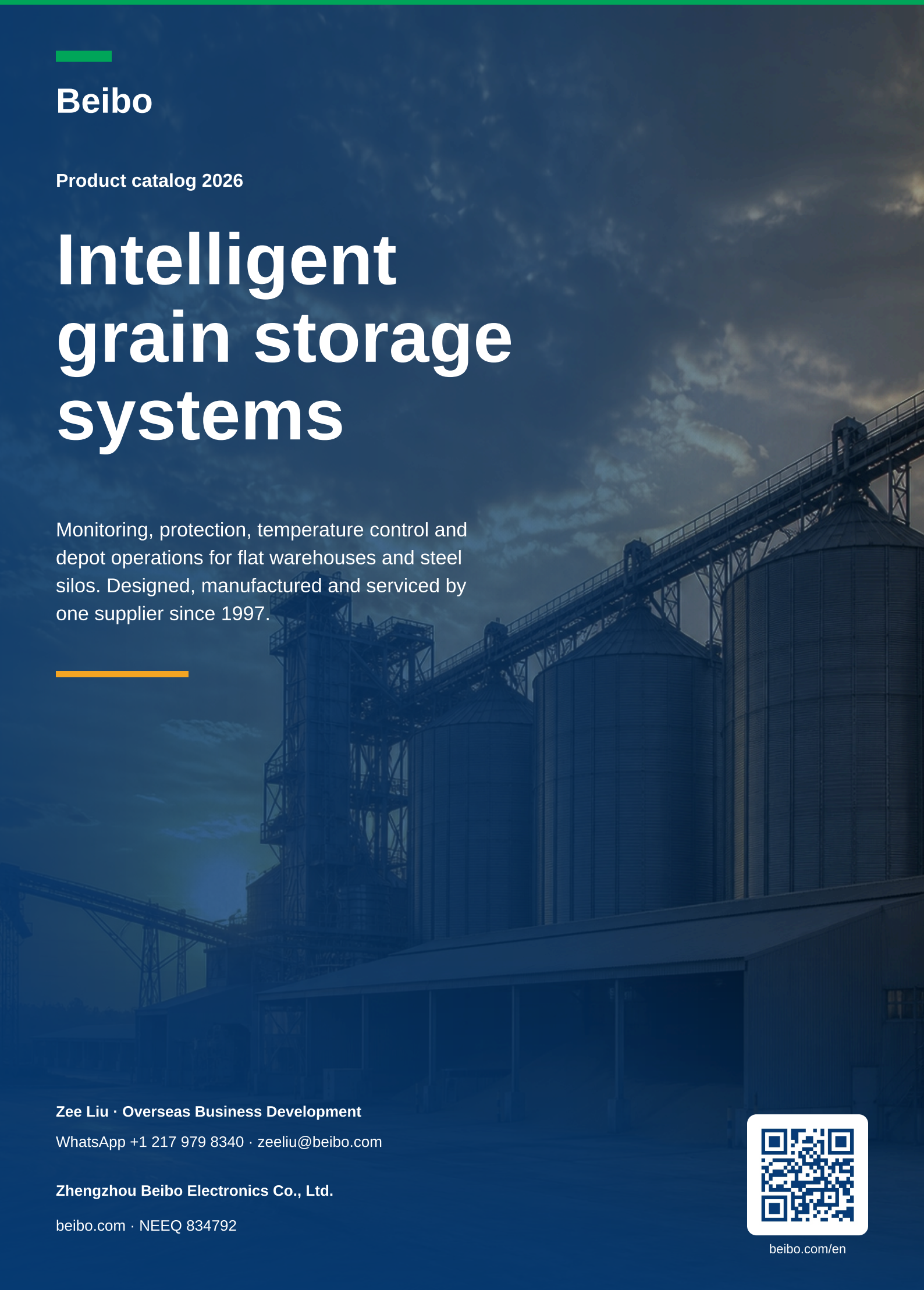
Beibo

Product catalog 2026

Intelligent grain storage systems

Monitoring, protection, temperature control and depot operations for flat warehouses and steel silos. Designed, manufactured and serviced by one supplier since 1997.



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beibo.com · NEEQ 834792

beibo.com/en

Five connected product systems

From grain condition to depot operations

Five systems cover sensing, control, protection and operations at the storage site. Each runs on its own; together they share one platform and one supplier.



Grain Condition Monitoring

Temperature, humidity, moisture, pest and gas detection; wired, wireless and solar field units



Temperature Control

Aeration control, motorised vents and windows, fans and grain-cooling units



Grain Protection

Recirculation fumigation, on-site phosphine generation, off-gas photodegradation, nitrogen controlled atmosphere



Inbound & Outbound

Vehicle entry, weighing, grain sampling and records at the depot gate



Storage Platform

One interface for depot-level visibility, equipment control and operations, on desktop and mobile

Any system can be supplied on its own. The integrated set shares one control cabinet, one platform and one support contact.

One platform for the entire depot

Smart Storage Management Platform

Live grain condition, equipment status and depot operations in one system, on desktop and mobile.



Monitoring · protection · control · operations

Storage visibility

Per-bin grain temperature, humidity and gas on a 3D depot view, with history per bin.

Equipment control

Remote start/stop and setpoints for aeration, cooling and fumigation equipment; status feedback from every device.

Operations records

Inbound and outbound quantities, sampling results, personnel and messages in one log.

Three views, one data set

Depot, storage condition and silo views

English-language interface. The views shown are from a deployed system; layout follows the site.



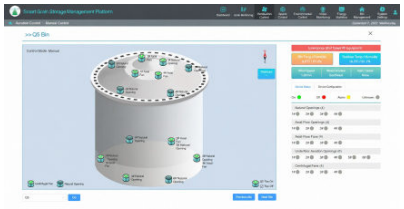
Depot overview

Storage locations and site-level operating status



Storage-condition dashboard

Grain, air and equipment data for one storage location



Silo condition view

Cable positions and readings for an individual silo

- **AI-assisted diagnostics**

  Alerts on rising grain temperature, gas concentration and pest activity; insect species identified from in-bin images.

- **Desktop and mobile**

  The same data set on the control-room workstation, a local terminal and a mobile phone; remote access over the internet.

Platform modules are enabled per site: monitoring, equipment control, inbound/outbound, access control and electronic storage cards.

Environmental monitoring

Know storage conditions at every point

Grain temperature, air temperature, humidity, moisture and gas across flat warehouses, shallow circular silos and vertical silos, from cable to control room.

- **Communications**

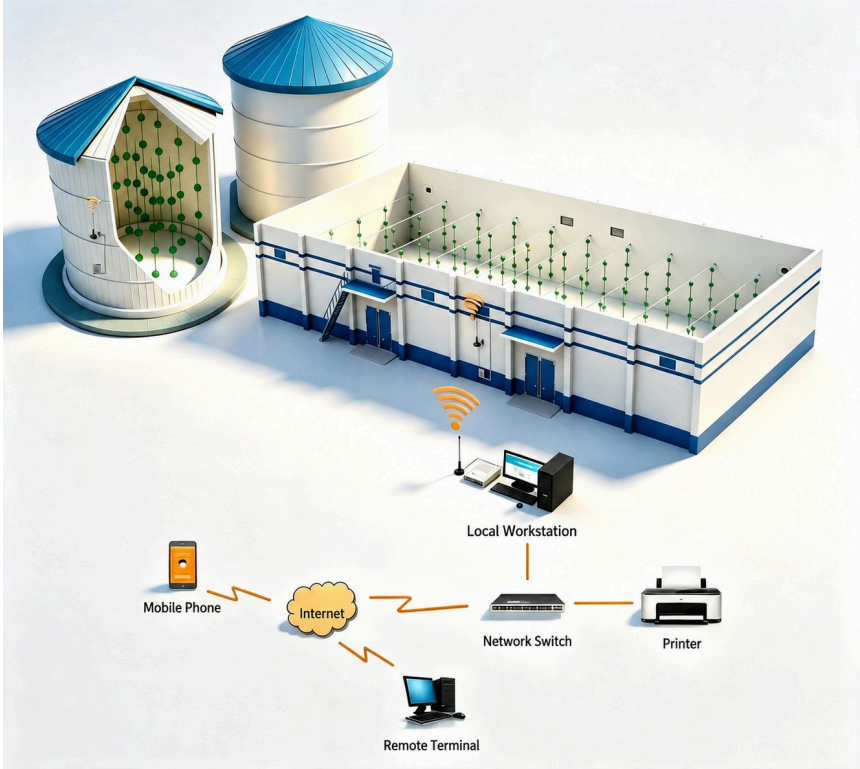
  RS-485, Ethernet or wireless between the field units and the control room; solar-powered units where there is no mains supply.

- **Access**

  Central control room, local workstation, mobile phone and remote terminal share one data set.

- **Structures**

  Flat warehouses, shallow circular silos and vertical steel silos; cable arrays or wireless probes per structure.



−40 to 60 °C
Air-temperature measurement range

10–99% RH
Air-humidity measurement range

±0.5 °C · ±3% RH
Temperature and humidity accuracy

Field units

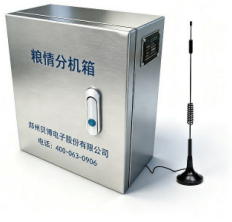
Grain-condition field units

Stainless-steel field units collect the cable readings in the bin and relay them to the control room. Wired, wireless and solar-powered versions cover sites with and without mains power.



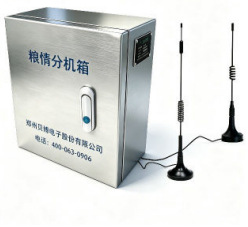
Wired field unit

RS-485 to the control cabinet



Wireless field unit

Radio link to the control room; no data cable between bins



Dual-antenna wireless unit

Two antennas for larger sites



Solar-powered wireless unit

No power or data cable to the bin



Acquisition board

Multi-channel cable inputs with DIP-switch addressing; one board per field unit



Handheld reader (dc11)

Reads and stores grain temperature on site; USB transfer to the platform. −55 to 125 °C, ±0.5 °C, sensor distance up to 100 m

512 + 6
Temperature and humidity sensors per controller

304 stainless
Enclosure; Zone 21 dust-atmosphere option

IP68
Portable collector and cable protection rating

Temperature cables

Cables for flat storage and silos

Digital and analogue temperature cables with 2–5 measuring points per cable, steel-wire load core, HDPE sheath and 300 kg tensile strength. Cable and portable collector rated IP68.





Lifting ring

Suspension from the silo roof



Flange feed-through

Sealed roof penetration



Bottom weight

Keeps the cable vertical in the grain

dc09 · Round cable, flat storage	Single-shot moulded, smooth surface, copper conductor; 2–5 points per cable, customisable
dc07 · Flat cable, flat storage	Two-shot moulded, removable sensors, steel-wire load core; 2–5 points per cable
dc08 · T-connection	Joins 2–6 cables without inline splices; one cable can be replaced without breaking the chain
dc05 · Digital round silo cable	Steel wires resist crushing and static; small cross-section reduces drag in the grain
dc04 · Digital flat silo cable	Adjustable tensile specification; several hanging arrangements
dc06 · Analogue silo cable	Resistive sensing, lightning and static protection, fumigant-sealed sensors; one failed point does not take the cable out of service

Air, humidity and moisture

Sensors and environmental controllers

Sensors for air temperature, humidity and grain moisture, and the controllers that switch the equipment correcting them.



Temperature & humidity sensor (kf03)

Sensirion digital element. −55 to 125 °C ±0.5 °C; 0–99 %RH ±2 %RH; ceiling mount, Ø107 × 37 mm



Multi-point temperature transmitter

Cable-array interface for steel silos and flat storage; digital signal on the shared communication line



Environmental controller (kf02)

−40 to 80 °C, 1–95 %RH; 16 sensors, 16 control channels; drives fans, windows and dehumidifiers

dc10 · Humidity sensor	HUMIREL HS1101 element; 0–100 %RH, ±5 %RH, 5 s response, 10-year service life; digital output on the temperature-cable line
Finished-grain store sensor	Colour display; 8–30 V DC; RJ45 / RS-485; Modbus RTU
kf01 · Regulation controller	Actuates equipment on instruction from the recording controller; AC 220 V, 10 A; wall mount, 198 × 91 × 42 mm
BB01 · Online moisture sensor	Curved electrodes; strain-gauge pressure and temperature/humidity compensation; calibrated to GB 5009.3-2016

Pest & gas detection

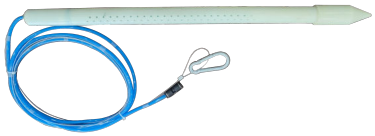
Detect conditions before a problem grows

Pest imaging, mobile detection hosts and multi-point gas sampling in one modular field system, sharing the temperature and humidity cables already in the bin.



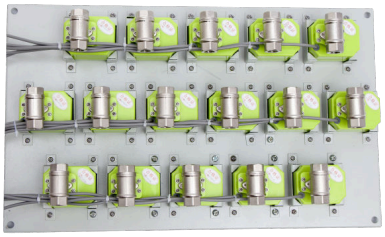
Mobile pest detection host (BB04)

Moves between bins; one host serves several storage structures



Pest imaging unit (BB05)

Samples insects at several points in the bin and photographs them on command; self-cleaning plate, snap-fit jar



Sampling selector module

Multi-point gas sampling through one analyser (invention patent)

Gas-monitoring ranges

PH_3
0–2,000 ppm · ±3% F.S.

O_2
0–30% · ±1% Vol.

CO_2
0–50,000 ppm · ±5% F.S.

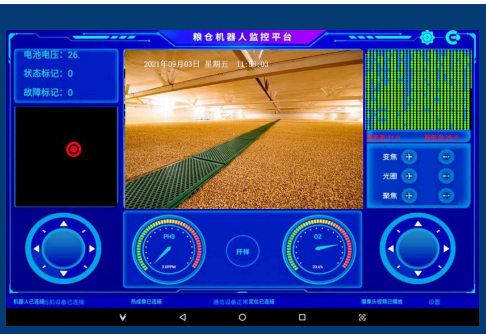
- **AI species identification**
  Insect species identified from in-bin images on the platform, ahead of manual sampling and laboratory results.
- **Integrated cabinet (BB03)**
  Temperature, humidity, pest and gas monitoring with control of aeration, recirculation, air-conditioning and controlled-atmosphere equipment in one cabinet.

Inspect without entering the grain

Grain-surface inspection robot

A tracked in-bin device for grain-surface temperature, gas detection, grain sampling and video monitoring, operated from outside the bin.





Four tasks in one field device

- Grain-surface temperature
- Gas detection
- Grain sampling
- Video monitoring

Operators stay outside

Bins that carry a gas or entrapment risk are inspected from the platform; the robot returns temperature, gas and video.

Surface sampling

Takes a grain sample from the surface for laboratory analysis without a person on the grain.

Operated with the pest and gas detection system; readings are logged to the same bin record as the cable data.

Aeration control

Automate the air path

Master panel, sub-control boxes, motorised openings and fans run aeration on grain and air temperature and humidity data, manually or automatically, with open/closed feedback from every device.



Electric rain cap

Closes automatically in rain or snow; opens for aeration



Motorised ventilation window (xx01)

Air-tightness half-life 60 s at 500 Pa; service life over 10 years



Motorised air damper

Worm-gear actuator, 0–120° opening



Axial fan (xx08)

826–68,000 m³/h; 38–474 Pa; 0.025–11 kW; explosion-proof variants

xx12 · Master control panel	Controls axial and centrifugal fans, vents and windows on grain and air temperature/humidity data; remote manual or automatic
xxx10 · Sub-control box	Opens and closes vents, windows and fans in remote or local mode; collects open/closed status from each device
xx04 · Fully electric vent	2 mm steel cover with polyurethane foam core; opens beyond 95° in 28 s; DC 24 V, 2 A; half-life 60 s at 500 Pa
xx03 · Axial fan window	Opens beyond 90° in 30 s; 90 W motor; micro-switch limit sensing; stainless-steel housing
BB02 · Electric actuator	0–120° opening; cast-aluminium RV worm gear; manual and remote control, cloud-capable

Field hardware for temperature control

Integrated temperature control

Control cabinets, motorised insulated windows and cooling units coordinate the air path at the bin, from one cabinet and one screen.



Grain cooling unit (kt01)

6HP grain-storage cooling unit; runs standalone on its own inlet and outlet temperatures



Shallow-silo control cabinet (xx09)

304 stainless steel, 1.5 mm; 640 × 320 × 1200 mm; 12-inch colour touchscreen; Ethernet; AC 220 V



Motorised outward-swing window

Insulated; AC 220 V, 40 W; closes in 20 s; air-tightness half-life 300 s at 500 Pa (utility model)

Cooling

Dedicated grain-cooling units, fumigation-compatible

Ventilation

Motorised windows, vents and fans with status feedback

Control

One cabinet for monitoring, equipment control and management

Every window, vent and fan reports its open/closed state to the platform, so the air path can be verified remotely.

Recirculating air temperature conditioning

Create a controlled internal air loop

A closed loop uses the existing bottom aeration ducts, a recirculation fan and insulated piping to move air inside the bin without exchange with outside air.







Duct intake

Draws air from the bottom aeration ducts

Insulated pipe

Returns the air to the upper storage space

Closed loop

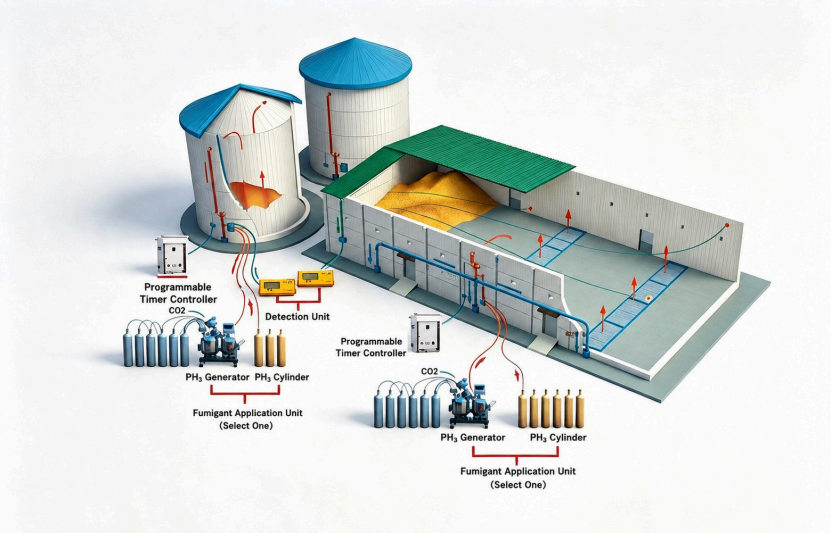
No exchange with outside air during circulation

Equipment list and specifications are prepared against storage type and capacity; contact us with your bin dimensions.

Recirculation fumigation

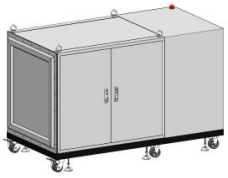
Fumigant distributed through the whole grain mass

Phosphine generated on site, driven through the bin by an explosion-proof recirculation fan, concentration monitored in the grain mass, off-gas photodegraded before discharge.





Phosphine generator (za02)



Off-gas photodegradation unit

za02 · Phosphine generator	Aluminium phosphide tablets or pellets react under controlled conditions; 1–8 kg per charge; 2–15 kPa working pressure; outlet ≤ 45 ± 5 °C; AC 220 V, 0.18 kW; 1600 × 800 × 1700 mm; 270 kg
za01 · Recirculation fan	0.75 kW explosion-proof; under 1,000 m³/h and 1,000 Pa; 2,800 rpm; 304 stainless housing, aluminium impeller, epoxy-coated casing
za03 · Mobile recirculation unit	For gas-tight structures with aeration ducts; 0.75 kW, 380 V; IP55; all pipework stainless steel
Off-gas photodegradation	Fumigation off-gas photodegraded before discharge; invention patent ZL 2009 1 0064940.6

Nitrogen controlled-atmosphere storage

Manage the storage atmosphere

A controlled-atmosphere control cabinet and a mobile nitrogen-generation unit hold a low-oxygen atmosphere in gas-tight bins, run from the same platform as monitoring and gas detection.



Controlled-atmosphere control cabinet



Mobile nitrogen-generation unit

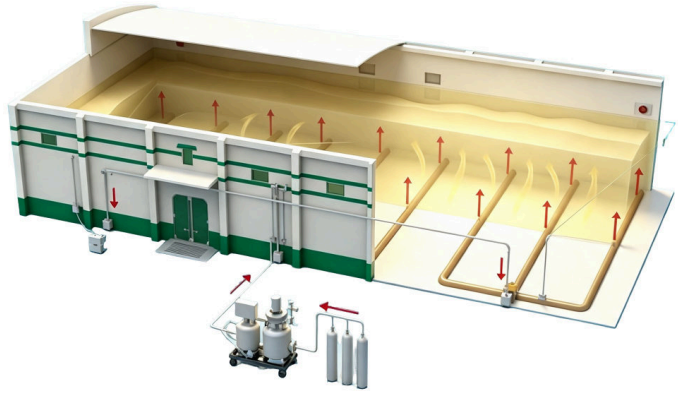
- **System equipment**
  Controlled-atmosphere control cabinet and mobile nitrogen-generation unit, sized to the bin volume.
- **Integrated deployment**
  Runs with oxygen and phosphine gas detection and the access-control system on one platform.
- **Site engineering**
  Gas-tightness, unit capacity and pipework are specified against the storage structure.

Supplied under SINOGRAIN centralized procurement alongside monitoring, fumigation and aeration systems.

Grain cooling

Actively cool the stored grain

Where ambient air cannot cool the grain, the 6HP grain-storage cooling unit (kt01) runs standalone against its own inlet and outlet temperatures, on the panel or remotely.





Grain cooling unit (kt01)



Unit and ducting arrangement

- **Compressor and coating**
  Variable-refrigerant-flow scroll compressor; imported anti-phosphine coating; fumigation-compatible.
- **Installation**
  Indoor unit mounted outside the bin; dedicated cold-air delivery fan for large-span bins.
- **Control**
  Start/stop, cooling setpoint, inside and outside temperature and status on the panel or from the platform; CCC-certified HLD series.

Supervisory software tracks grain temperature and humidity against a low-temperature storage model and schedules cooling runs.

Make every load visible

Inbound & outbound management

Vehicle entry, weighing, grain sampling and records at the depot gate, written to the same platform as the storage data.





Operations workflow

Vehicle registration at the gate
Weighbridge in and out
Grain sample inspection
Inventory and record update

- **Gate to inventory**
  Each load is tied to a vehicle, a weighbridge ticket, a sample result and a storage location.
- **Depot-level visibility**
  Inbound and outbound quantities appear on the platform alongside grain condition and equipment status.

Weighbridge, sampling and gate hardware are specified per site; records are kept per load and per storage location.

Smart access control

Make warehouse entry safer and traceable

Identity verification, gas awareness and entry records at every warehouse door, linked to warehouse lighting and the platform.



Access functions

- Face, fingerprint and password authentication
- Oxygen and phosphine concentration at the door, with alerts
- Entry / exit headcount
- Linked warehouse lighting and access records

3 methods
Face, fingerprint, password

O_2 · PH_3
Gas concentration checked before entry

Headcount
Entry / exit statistics per warehouse

Electronic storage cards at each bin show the same grain-condition and inventory data as the platform.

Integrated control equipment

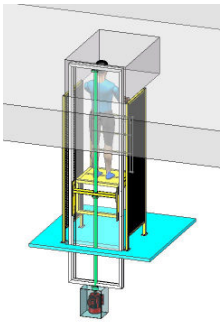
Bring subsystems into one control layer

Control cabinets and auxiliary equipment connect monitoring, detection and storage-control systems into one installation.





Electric manhole



Electric lifting platform

- **Integrated monitoring & control cabinet (BB03)**
  Real-time temperature, humidity, pest and gas monitoring with live display; controls aeration, recirculation, air-conditioning and controlled-atmosphere equipment.
- **Auxiliary equipment**
  Electric manholes and lifting platforms for safe access and inspection in shallow circular silos.

System integration: monitoring · gas · pest · aeration · temperature conditioning · controlled atmosphere · access control

A long-term partner for grain storage

Engineering expertise since 1997

Zhengzhou Beibo Electronics designs, manufactures and services grain-storage monitoring, control and management systems. Incorporated July 2003; listed on NEEQ in December 2015 (834792); National High-Tech Enterprise.


Grain-surface robot, flat warehouse


Recirculation fumigation install


Temperature cables in bagged storage


Depot gate, inbound weighing

Since

1997

Grain-storage expertise

Incorporated 2003
NEEQ stock code 834792
National High-Tech Enterprise
SINOGRAIN approved supplier

7,000+
Installation points across China

15
Provinces in the partial customer directory

22 · 26
Patents · software copyrights

- **National and provincial reserves**
  Central Grain Reserves Corporation depot companies; provincial storage and reserve operators.
- **Local and commercial storage**
  Municipal and county reserves; grain, grain-and-oil and storage-management companies.

Customer directory shown as an indicative, partial list; individual customer names are available on request.

Credentials that support delivery

Company and product qualifications

Management-system certifications, enterprise credentials and product compliance. Certificate numbers and validity dates are available on request.



ISO 9001 quality management



ISO 14001 environmental management



Occupational health and safety management



National High-Tech Enterprise



Software enterprise certificate



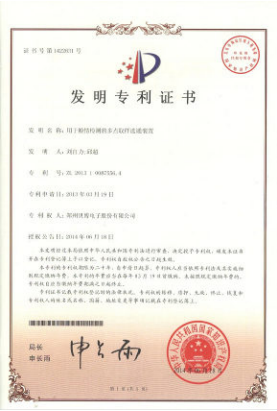
CCC product certification (HLD-series cooling units)

Representative records from the company and product qualification archive.

Intellectual property across hardware and software

Patent and software-rights records

22 patents and 26 software copyrights across monitoring, sampling, fumigation, ventilation and storage-management software.



Multi-point sampling selector

Invention patent



Wireless sensor technology

Invention patent



Phosphine off-gas photodegradation device

Invention patent



Integrated multi-parameter detection rod

Invention patent



Motorised outward-swing insulated window

Utility model



Smart storage management software

Software copyright

Selected records from the invention-patent, utility-model and software-copyright archive.

Product index

Product codes and online specifications

Every product in this catalogue has a specification page on beibo.com. Scan the code for the system, or send the product code with your enquiry.

Grain condition monitoring

dc04 · dc05 · dc06 · dc07 · dc08 · dc09 cables · dc10 humidity sensor · dc11 handheld reader · kf01 · kf02 · kf03 controllers and sensors

Pest & gas detection

BB01 online moisture sensor · BB02 electric actuator · BB03 integrated cabinet · BB04 mobile pest host · BB05 pest imaging unit

Aeration control

xx01–xx05 windows and vents · xx06 mixed-flow fan · xx07 centrifugal fan · xx08 axial fan · xx09 shallow-silo cabinet · xxx10 sub-control box · xx12 master panel

Grain cooling

kt01 6HP grain-storage cooling unit

Recirculation fumigation

za01 recirculation fan · za02 phosphine generator · za03 mobile recirculation unit · off-gas photodegradation unit

Recirculating air conditioning

Equipment list prepared against storage type and capacity

Smart storage management platform

Monitoring · equipment control · inbound/outbound · access control · electronic storage cards

Codes as listed on beibo.com/en/products at the time of printing.

Beibo

Intelligent grain storage systems

Let’s plan your grain storage system.

Send us your storage type, capacity and existing equipment. Our engineers come back with a system configuration and an outline proposal.

Zee Liu

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Grain monitoring · Pest control · Aeration · Temperature conditioning · Grain cooling · Fumigation · Platform management

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